

Feasibility by Physics, Not by Penalty

Penalty-Free Hybrid Quantum Portfolio Construction

WISER Global Quantum+AI 2026 · Vanguard Challenge

Multi-asset allocation copilot: HRP decomposition · ADMM coordination · constraint-preserving QAOA+ · convex polish

The Problem: Exactly K of N, Inside Hard Guardrails

Master problem (binary z , continuous w)

$$\min w' \Sigma w - \lambda_{\text{ret}} \mu' w + c' |w - w_{\text{prev}}| + w_{\text{cvar}} \cdot \text{CVaR}_{\alpha}(-Rw)$$

$$\text{s.t. } 1'w = 1 \quad \cdot \quad 0 \leq w \leq W_{\text{max}} \cdot z \quad \cdot \quad w \geq w_{\text{min}} \cdot z$$

$$1'z = K \text{ (cardinality)} \quad \cdot \quad z \in \{0,1\}^N$$

$$\text{class caps } S \cdot w \leq c \quad \cdot \quad \text{income } y'w \geq \text{floor}$$

Quadratic objective, linear constraints, binary decisions — the cardinality constraint makes it NP-hard.

$$5.4 \times 10^{20}$$

feasible portfolios at $N=100, K=20$ — $C(100,20)$

10 classes

multi-asset universe: equities, bonds, high yield, commodities, REITs, alts

Synthetic multi-asset data: returns, volatilities, hierarchical correlation, transaction costs, income yields — all seeded and reproducible.

Architecture: Four Pillars, One Handshake

1 · HRP Decomposition

Ward clustering on correlation distance carves $N=100$ into 9 clusters (≤ 20 assets). 96% class purity. Budgets $\propto$ inverse cluster risk.



2 · ADMM Coordinator

$x \rightarrow z \rightarrow u$ loop per cluster with adaptive ρ (residual balancing, $\mu=10, \tau=2$). Converged 9/9 clusters to residual $\sim 10^{-5}$.



3 · Classical x-step

CVXPY convex QP: risk, return, L1 transaction costs + ADMM anchor. Certifiable optimum every iteration.



4 · Quantum z-step

Penalty-free QAOA+ on HRP-compiled tree mixers. ADMM term folds into the Ising field — linear in z , zero extra gates.

Then a global convex polish: free weights on the union support against the full covariance, with all guardrails and all four investor goals active. The NP-hard combinatorics live in the quantum layer; everything continuous stays convex and certifiable.

The Core Claim: The Constraint Is a Symmetry of the Circuit

Textbook approach adds a penalty: $\lambda(\mathbf{1}'\mathbf{z} - K)^2$ — a knob that must be tuned, and every mis-tuning either leaks the constraint or drowns the market signal.

We delete the penalty entirely.

- Start in the Dicke state — uniform superposition over ALL weight-K portfolios
- Mix with XY blocks $e^{-i\beta(XX+YY)/2}$ — every gate conserves the number of selected assets
- Phase-separate with the pure mean-variance Hamiltonian — couplings are market structure, nothing else

10^{-33}

measured probability of an illegal portfolio, at random circuit parameters

100%

feasible shots — for every γ, β , every depth, Trotterization included

No λ to tune. No feasibility repair. No wasted shots. Verified numerically, not asserted.

Compiler-Level Design: Topology-Matched Tree Mixers

The mixer graph is not generic — it is compiled from the market.

Per cluster, the classical layer emits the maximum-correlation spanning tree (BFS-ordered, degree capped at 3 = the heavy-hex lattice degree). XY excitation hops travel between the closest economic substitutes first, and the tree embeds on the chip without SWAP routing.

Connectivity keeps every weight-K portfolio reachable; the degree cap turned out to be free — mixer depth dropped to 3 parallel two-qubit layers, matching a ring.

$\text{deg} \leq 3$

every mixer tree embeds on heavy-hex — by construction, all 9 clusters

3 layers

parallel two-qubit slabs per mixer application (was 4–8 uncapped)

Note: the Dicke prep + XY mixer cost real gates — at matched size our circuit is bigger than a penalty circuit. The win is architectural (next slides).

Tunable Investor Goals — Each With a Measured Effect

GROWTH

λ_{ret} return emphasis

Sweeps the risk-return trade-off in both layers; frontier traced live in the copilot.

INCOME

$y'w \geq \text{floor}$

2.5% floor $\rightarrow$ book delivers 2.68% yield. Unattainable floors relax gracefully — flagged, never silent.

DRAWDOWN

CVaR_ α scenario penalty

Worst-15% monthly scenario loss cut 6.26% $\rightarrow$ 5.23% (Rockafellar–Uryasev, convex).

COST

t-cost multiplier

Per-asset linear costs scale up to 5 $\times$; optimizer holds the previous book harder.

All four goals act on the convex capital layer — they never touch the quantum circuit, so feasibility stays guaranteed while goals move.

Measured Outcomes (N=100, K=20, seeded — reproducible)

0

hard-guardrail breaches — guaranteed by construction, confirmed by audit

0.47

Sharpe, net of transaction costs

1.0000

approximation ratio vs certified exhaustive optimum

34 s

full pipeline on a laptop, quantum firing in 9/9 clusters

Validation against classical routines (challenge requirement)

Exhaustive enumeration of the $C(n,K)$ shell (certified optimum) · branch-and-bound MIQP · cardinality-preserving simulated annealing · full-universe HRP.

vs HRP baseline: HRP diversifies over all 100 names — but structurally violates the K-name mandate (1 breach). OMEGA delivers the mandated exactly-20 book at comparable risk-adjusted quality with zero breaches. ADMM residuals decayed 9/9 clusters to $\sim 10^{-5}$; adaptive ρ activated in 9/9; ZNE cut noisy-energy error $6\times (0.0013 \rightarrow 0.0002)$.

Adversarial Benchmark: We Ran Their Stack, Correctly

Full-strength replication of the DBSCAN + soft-penalty-QUBO + X-mixer pipeline — encoding verified exact against brute force. Failure modes are measured, never scripted.

$\lambda = 0.05$ (small penalty)

2.3% feasible shots. Most probable outcome: BUY NOTHING — an empty portfolio.

$\lambda = 5$ (large penalty)

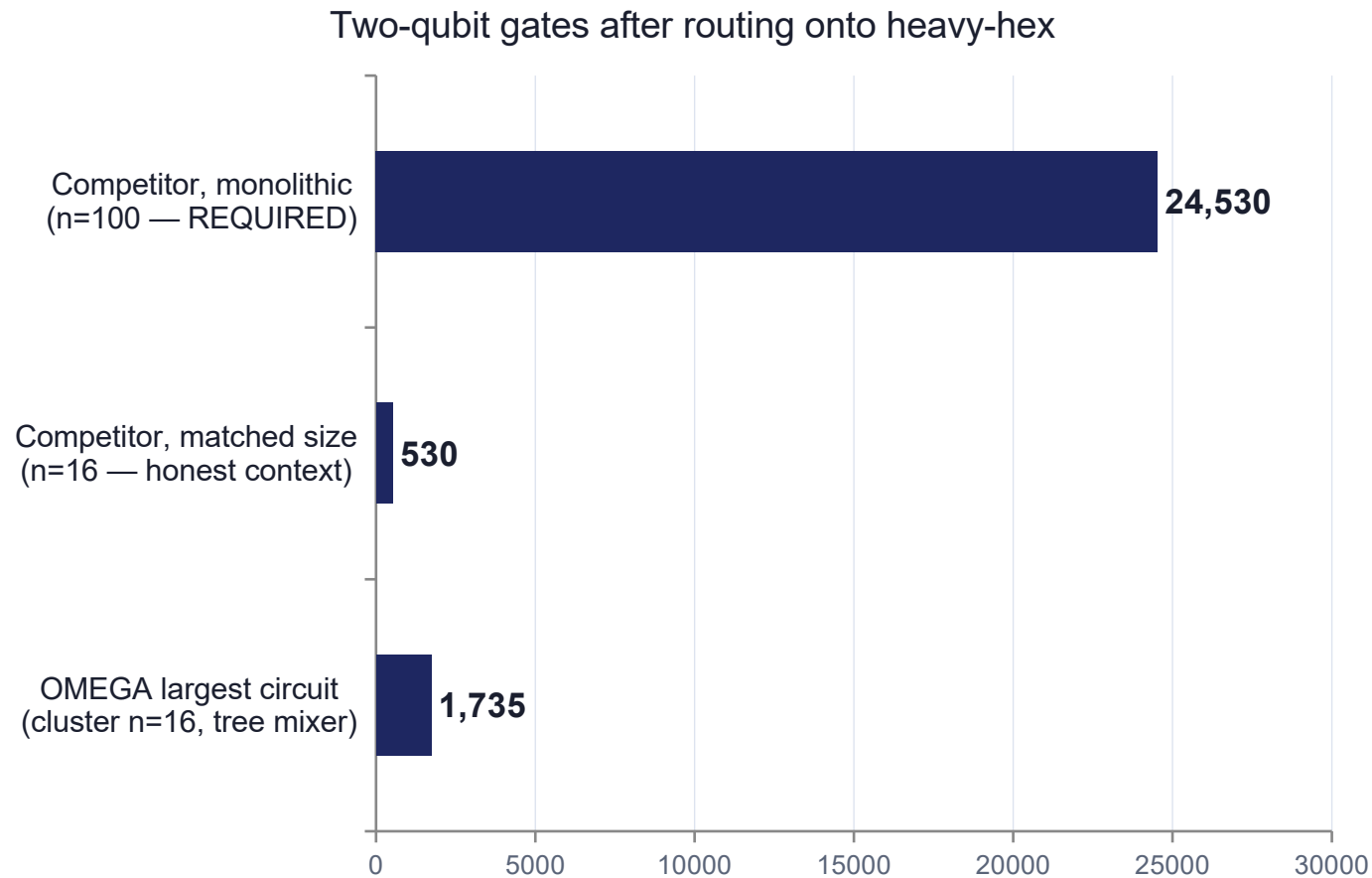
Feasible — but market signal rides on 4.9% of each rotation angle: below realistic pulse calibration.

DBSCAN at defaults

Labels ALL 100 assets as noise. No dendrogram to cut at a qubit budget. HRP: 96% purity, 0 discarded.

The double bind, demonstrated: no λ is right. Small λ leaks the mandate; large λ erases the market. And because the penalty couples all N assets, their formulation cannot be decomposed — whatever λ they pick, they must run one monolithic N -qubit circuit.

Hardware Audit: Real Transpiler, Real Heavy-Hex Lattice



14,630

SWAP-tax CX gates the monolithic penalty circuit pays for routing alone

14×

more noise exposure than OMEGA's largest circuit — same N=100 problem

At matched size their circuit is cheaper (530 vs 1,735) — we show it. The point is architectural: they can never run small; we never need to run big.

Scope & Roadmap

Built for the hardware we're heading toward

- Every circuit ≤ 20 qubits — sized for the real NISQ devices of today, not a distant fault-tolerant future.
- Compliance gate and per-run audit trail packaged as a deployable engine — not just a notebook.
- Directly relevant: every institutional fund must hold a fixed set of names under hard mandates.

The path to practice

- Execute on a real QPU — circuits are already compiled to the heavy-hex layout, SWAP-free.
- Scale to cluster sizes where classical enumeration becomes intractable — where quantum earns its keep.
- Integrate a professional risk model and live data feeds for production deployment.

At today's scale the advantage is structural — a hard compliance guarantee and a hardware-ready architecture, built to earn quantum advantage as problem sizes grow.

**Feasibility is not a penalty term.
It's a symmetry.**

Multi-asset allocation copilot: guaranteed-compliant, hardware-ready, every number measured.

github.com/hastikacheddy/wiser-vanguard-quantum